

Assignment: Practicing MongoDB CRUD & Aggregation:

Perform MongoDB Tasks Step by Step

Task 1: Insert Documents Insert at least 7 employee records with the following fields:

```
< switched to db employee
> db.createCollection("Faculty")
< { ok: 1 }
> db.Faculty.insertMany([{"name":"Irfan",eid:101,age:22,department:"IT",city:"Mum
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('68e5f1413b24fe8c6b165d42'),
    '1': ObjectId('68e5f1413b24fe8c6b165d43')
  }
}
> db.Faculty.insertMany([{"name":"Irfan",eid:101,age:22,department:"IT",city:"Mum
< {
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('68e5f6343b24fe8c6b165d44'),
    '1': ObjectId('68e5f6343b24fe8c6b165d45'),
    '2': ObjectId('68e5f6343b24fe8c6b165d46'),
    '3': ObjectId('68e5f6343b24fe8c6b165d47'),
    '4': ObjectId('68e5f6343b24fe8c6b165d48'),
    '5': ObjectId('68e5f6343b24fe8c6b165d49'),
    '6': ObjectId('68e5f6343b24fe8c6b165d4a')
  }
}
```

Task 2: Basic Queries Find all employees:

```
> db.Faculty.find()
< {
  _id: ObjectId('68e5f1413b24fe8c6b165d42'),
  name: 'Irfan',
  eid: 101,
  age: 22,
  department: 'IT',
  city: 'Mumbai',
  salary: 20000,
  designation: 'Data Analyst'
}
```

Find employees who work in the "IT" department:

```
> db.Faculty.find({department:"IT"})
< {
  _id: ObjectId('68e5f1413b24fe8c6b165d42'),
  name: 'Irfan',
  eid: 101,
  age: 22,
  department: 'IT',
  city: 'Mumbai',
  salary: 20000,
  designation: 'Data Analyst'
}
{
  _id: ObjectId('68e5f6343b24fe8c6b165d44'),
  name: 'Irfan',
  eid: 101,
  age: 22,
  department: 'IT',
  city: 'Mumbai',
  salary: 20000,
  designation: 'Data Analyst'
}
```

Find employees with salary greater than 30,000:

```
> db.Faculty.find({salary:{$gt:30000}})
< {
  _id: ObjectId('68e5f6343b24fe8c6b165d47'),
  name: 'Madeeha',
  eid: 104,
  department: 'BCOM',
  city: 'Thane',
  salary: 35000,
  designation: 'Chartered Accountant'
}
{
  _id: ObjectId('68e5f6343b24fe8c6b165d4a'),
  name: 'Shahbaz',
  eid: 107,
  age: 25,
  department: 'Testing',
  city: 'Lucknow',
  salary: 40000,
  designation: 'Digital Marketing'
}
```

Find one employee from "HR" department:

```
> db.Faculty.find({department:"HR"})
< {
  _id: ObjectId('68e5f6343b24fe8c6b165d48'),
  name: 'Abuzar',
  eid: 105,
  age: 23,
  department: 'HR',
  city: 'Mysore',
  salary: 22000,
  designation: 'Intern'
}
```

Task 3: Updates:

Update salary of "Yash Diwate" to 60000:

```
> db.Faculty.update({name:"Yash"},{$set:{salary:60000}})
< DeprecationWarning: Collection.update() is deprecated
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.find({name:"Yash"})
> TypeError: db.find is not a function
> db.Faculty.find({name:"Yash"})
< {
  _id: ObjectId('68e5f6343b24fe8c6b165d45'),
  name: 'Yash',
  eid: 102,
  age: 24,
  department: 'CS',
  city: 'Chennai',
  salary: 60000,
  designation: 'Software Engineer'
```

Change designation of "Abuzar" to "Senior Developer" :

```
> db.Faculty.update({name:"Shahbaz"},{$set:{designation:"Senior Developer"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Faculty.find({name:"Shahbaz"})
< {
  _id: ObjectId('68e5f6343b24fe8c6b165d4a'),
  name: 'Shahbaz',
  eid: 107,
  age: 25,
  department: 'Testing',
  city: 'Lucknow',
  salary: 40000,
  designation: 'Senior Developer'
}
```

Update city of "Abuzar" to "Pune":

```
> db.Faculty.update({name:"Abuzar"},{$set:{city:"Pune"}})
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
> db.Faculty.find({name:"Abuzar"})
< {
  _id: ObjectId('68e5f6343b24fe8c6b165d48'),
  name: 'Abuzar',
  eid: 105,
  age: 23,
  department: 'HR',
  city: 'Pune',
  salary: 22000,
  designation: 'Intern'
}
```

Task 4: Deletions

Delete one employee with a salary less than 30,000:

```
> db.Faculty.deleteMany({salary:{$lte:30000}})
{
  acknowledged: true,
  deletedCount: 4
}
> db.Faculty.find()
{
  _id: ObjectId('68e5f6343b24fe8c6b165d45'),
  name: 'Yash',
  eid: 102,
  age: 24,
  department: 'CS',
  city: 'Chennai',
  salary: 60000,
  designation: 'Software Engineer'
}
```

Delete all employees from "Senior Developer" designation:

```
> db.Faculty.deleteOne({designation:"Senior Developer"})
{
  acknowledged: true,
  deletedCount: 1
}
> db.Faculty.find()
{
  _id: ObjectId('68e5f6343b24fe8c6b165d45'),
  name: 'Yash',
  eid: 102,
  age: 24,
  department: 'CS',
  city: 'Chennai',
  salary: 60000,
  designation: 'Software Engineer'
}
{
  _id: ObjectId('68e5f6343b24fe8c6b165d47'),
```